

PROJECT REPORT

On

(IOT) NOISE POLLUTION MONITORING SYSTEM

Submitted for the partial fulfilment of the award of

Master of Computer Application

Submitted by:

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(2024-2025)

Declaration

I hereby declare that the project work entitled Noise pollution Monitoring is an authenticated work carried out by me at Govt. College for Girls, Sector-14, Gurugram and this work has not been submitted for similar purpose anywhere else.

Date: 30 ARIL 2024

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Place: GURUGRAM

Enroll. No. : 2231914032

Class: MCA Ist year 2nd semester

Certificate

This is to certify that Sandhya Chauhan roll no. 2231914032 has worked Under my supervision to prepare her project report. The work embodied in this Report is original and was conducted at Govt. College for Girls, Gurugram. She Has completed all requirements of MCA ordinance.

Mrs. SARITA SHEERA

Associate Professor

(PROJECT GUIDE)

GCG, SEC 14, GURUGRAM, HARYANA

Acknowledgement

I would like to express my special thanks of gratitude to my teacher and project Guide Mrs. SARITA SHEERA for his valuable guidance. I am also thankful to all the Faculty members of the Comp. Sc. Dept. for their encouragement and help while Collecting information about the project.

Student name: SANDHYA CHAUHAN

Enroll no.: 2231914032

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INTRODUCTION

The system proposed in this is an advanced solution for monitoring the environmental conditions at a particular place and make the information visible anywhere in the world. The technology behind this is Internet of Things (IOT), which is an advanced and efficient solution for connecting the things to the internet and to connect the entire world of things in a network. Here things might be whatever like electronic gadgets, sensors and automotive electronic equipment. The system deals with monitoring and detection of sound pollution and sends sensor data to the computer and android phone using IOT link. The data updated from the implemented system can be accessible in the internet from anywhere in the world using TINKERCAD platform.

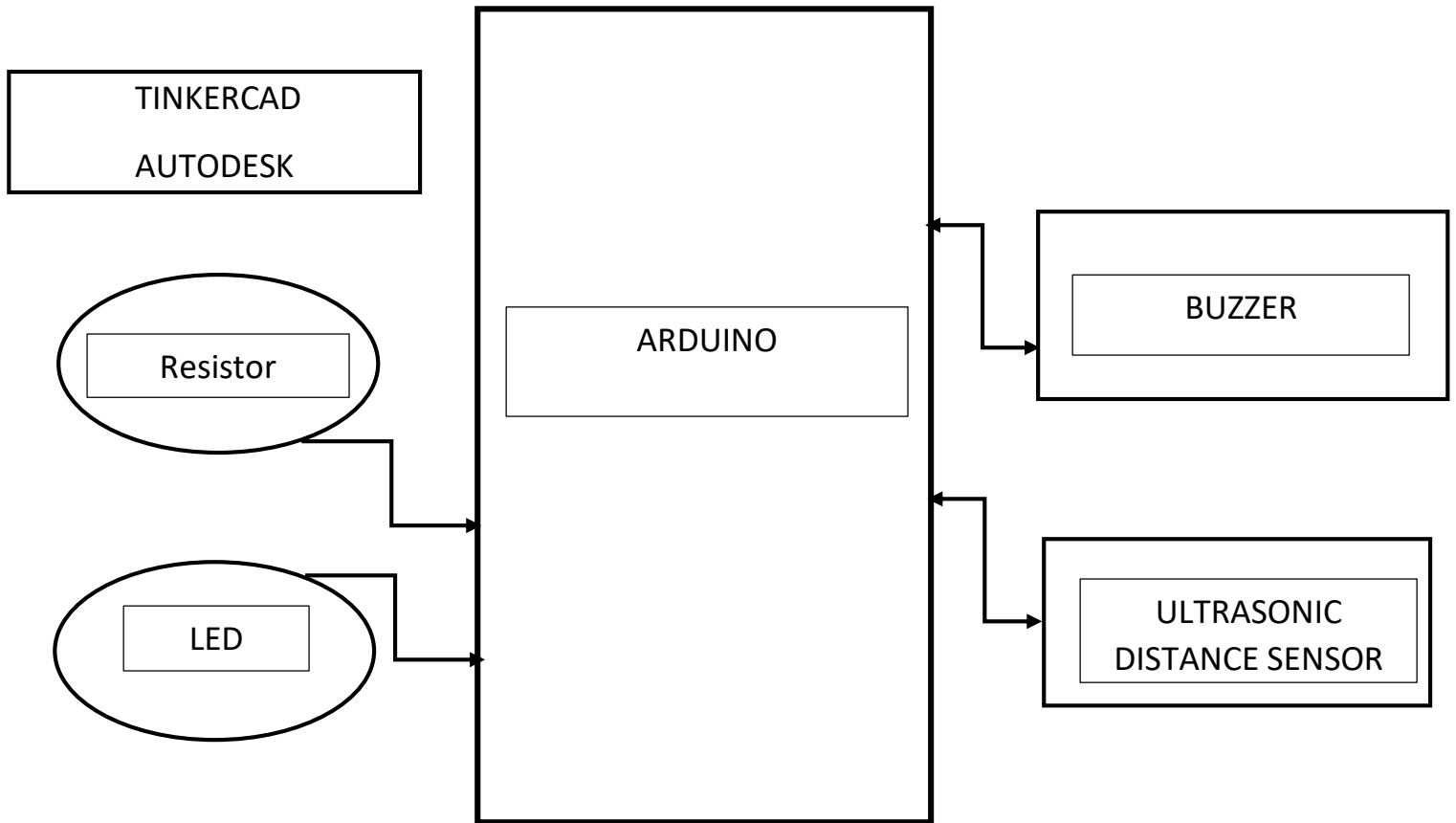
PROBLEM DEFINITION

OBJECTIVE

Objective of this project is to design and develop IOT noise pollution. The system consists of sound sensor. And send sensors data to the computer using IOT link. User can watch on thinkercad. The system will alert if any sensors reading goes above pre-set level. The system proposed in this is an advanced solution for monitoring the environmental conditions at a particular place and make the information visible anywhere in the world. The technology behind this is Internet of Things (IOT), which is an advanced and efficient solution for connecting the things to the internet and to connect the entire world of things in a network. Here things might be whatever like electronic gadgets, sensors and automotive electronic equipment. The system deals with monitoring and detection of sound pollution and sends sensor data to the computer and android phone using IOT link. The data updated from the implemented system can be accessible in the internet from anywhere in the world using TINKERCAD platform.

- To study the existing system.
- To design the block diagram.
- To decide the components specification & device in system.
- To design the circuit diagram and simulate it using suitable software.
- To design the PCB and implement hardware. To test the circuit and observe the result.
- To prepare report.

BLOCK DIAGRAM



REQUIREMENT

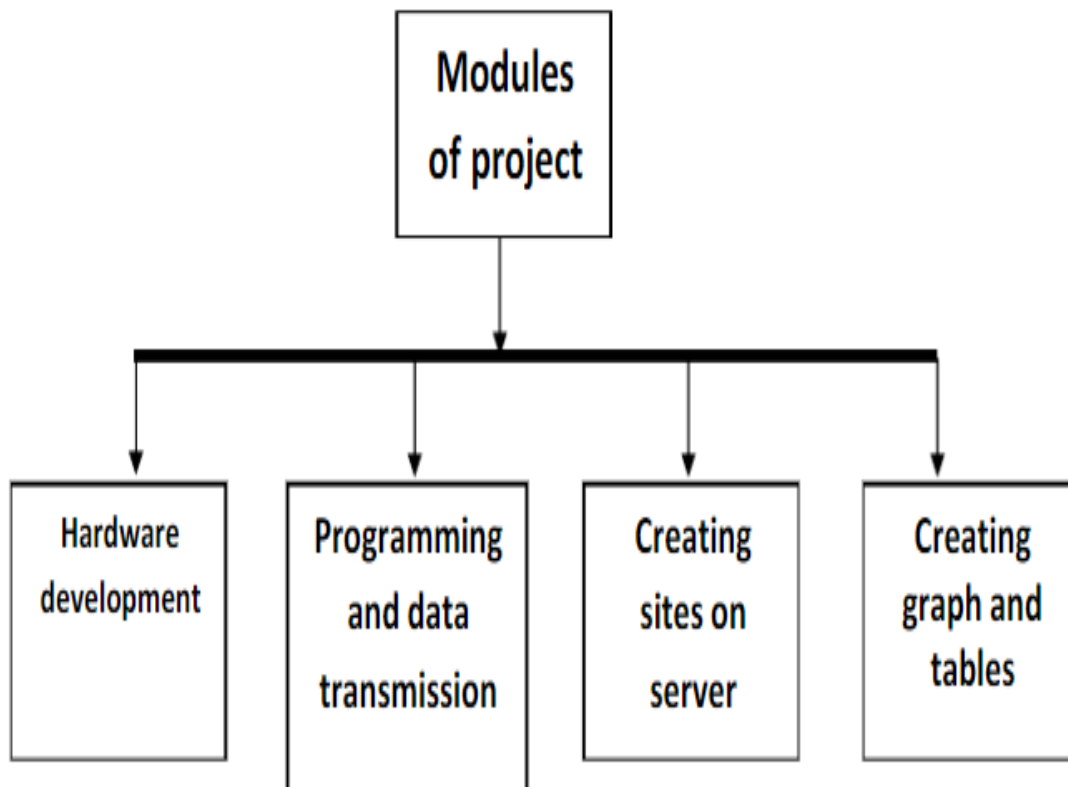
Hardware Used:

- Piezo buzzer
- Arduino Uno R3
- Ultrasonic Distance Sensor
- 1 k Ω Resistor
- Red LED
- 180 Ω Resistor
- 100 Ω Resistor
- LED RGB
- Bread Board

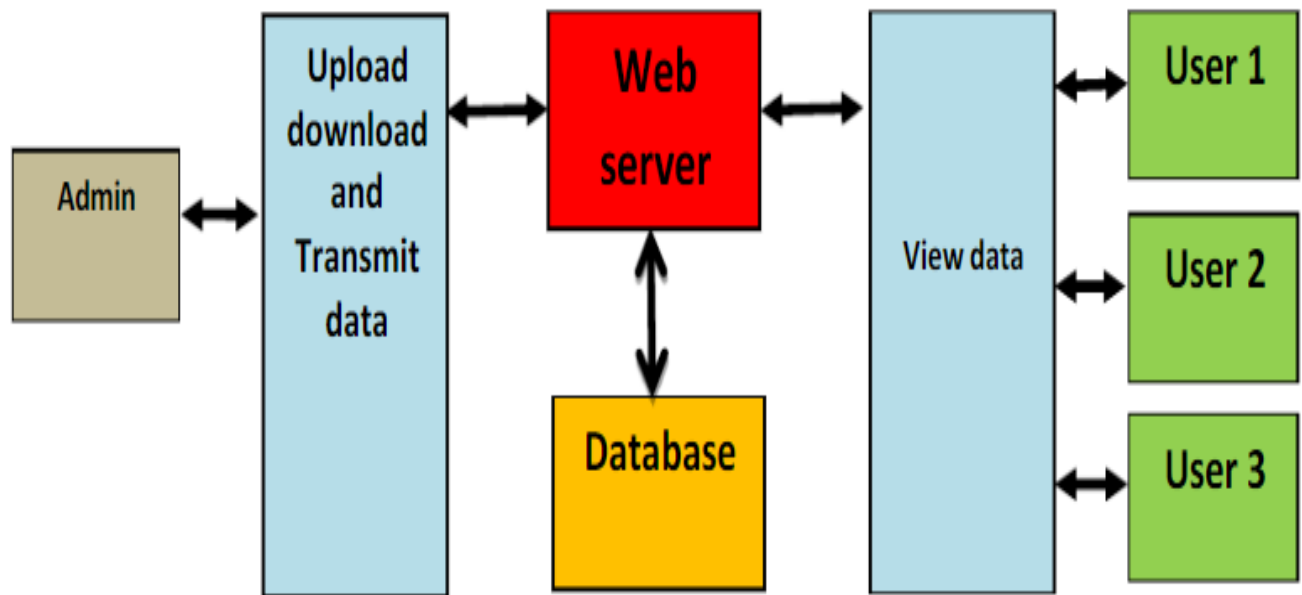
Software

- Arduino IDE programming microcontroller
- Tinkercad Platform

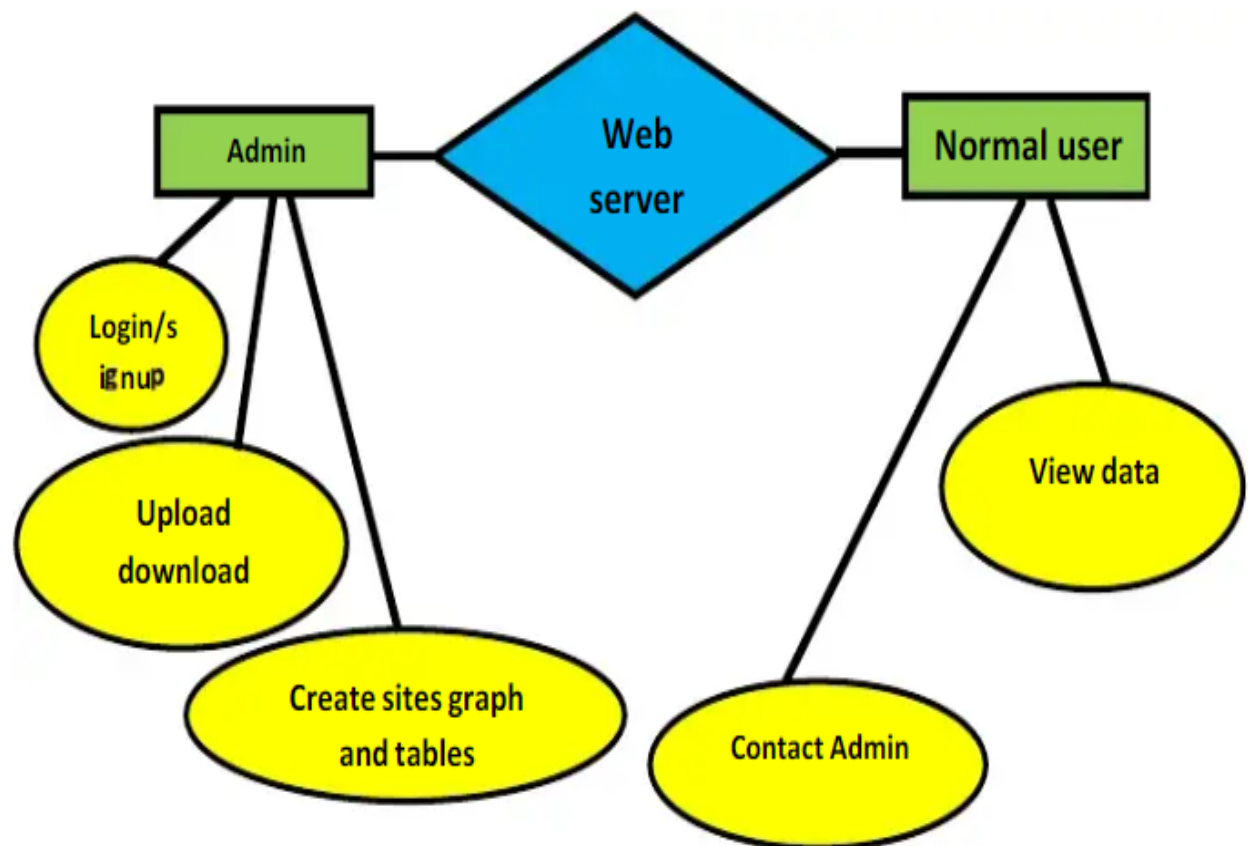
MODULES OF THE PROJECT



DATA FLOW DIAGRAM



E R DIAGRAM



Arduino Microcontroller

Arduino is an open-source electronics platform based on easy-to-use hardware and software. [Arduino boards](#) are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the [Arduino programming language](#) (based on [Wiring](#)), and [the Arduino Software \(IDE\)](#), based on [Processing](#).

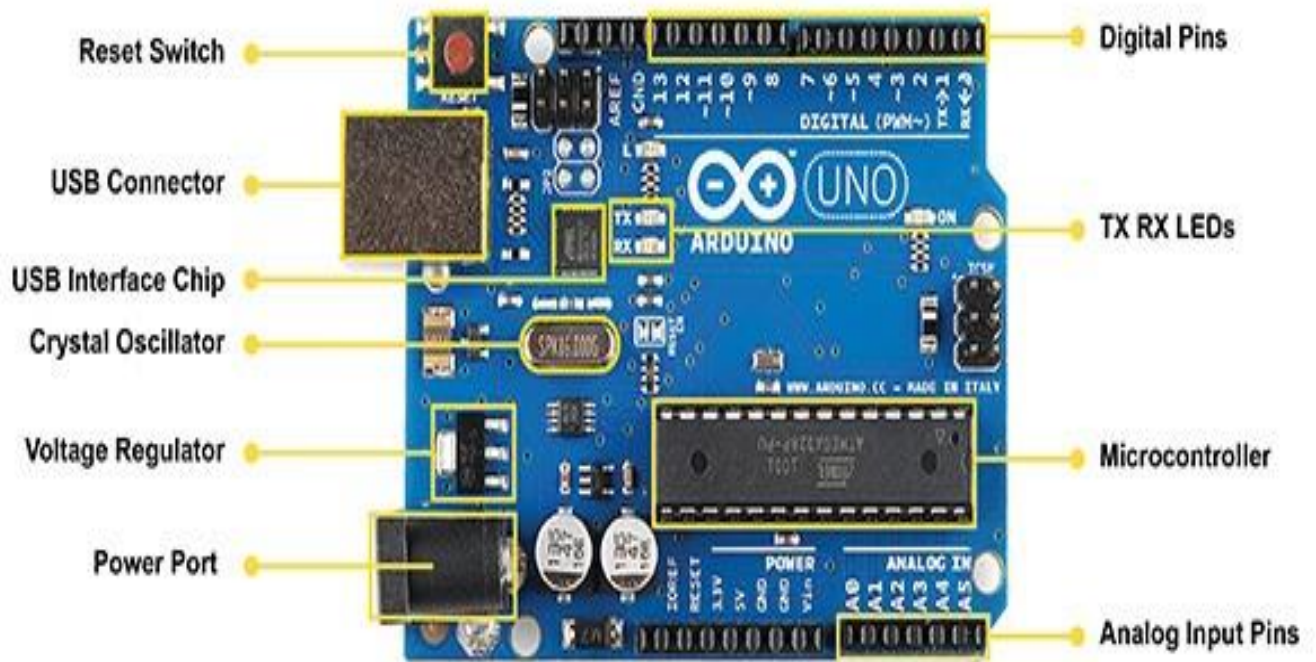
Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of [accessible knowledge](#) that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments.

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics. Designers and architects build interactive prototypes, musicians and artists use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things. Anyone - children, hobbyists, artists, programmers - can start tinkering just following the step by step instructions of a kit, or sharing ideas online with other members of the Arduino community.

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

ARDUINO:



Arduino Ultrasonic distance sensor

The Ultrasonic sensor or HC-SR04 is used to measure the distance of the object using SONAR.

It emits the Ultrasound at a frequency of 40KHZ or 40000 Hz. The frequency travels through the air and strikes the object on its path. The rays bounce back from the object and reach back to the module.

The four terminals of HC-SR04 are VCC, TRIG, ECHO, and GND. The voltage supply or VCC is +5V. We can connect the ECHO and TRIG terminal to any of the digital I/O pin on the specific Arduino board.

The Ultrasonic sensors work best for medium ranges.

The resolution is **0.3cm**.

The medium ranges of the sensor are **10cm** to **3m**. It works best at this duration.

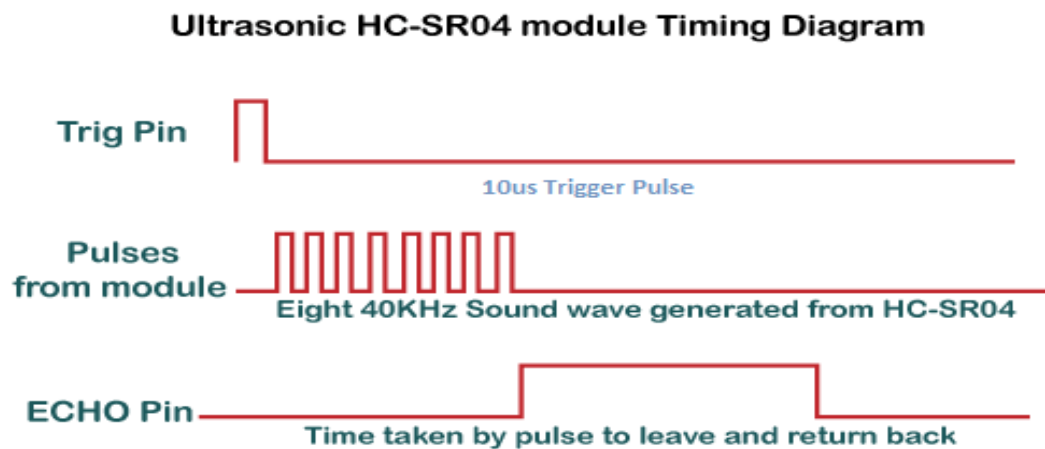
The maximum range the sensor may detect is **4.5m**.



Ultrasonic sensor Timing Diagram

We will set the TRIG pin to HIGH for some time (about 3 to 100 microseconds). As soon the TRIG pin is LOW, the Ultrasonic sensor sends the pulses and sets the ECHO pin to HIGH. When the sensor gets the reflected pulses, it sets the ECHO pin to LOW. We need to measure the time for which the ECHO pin was HIGH.

The timing diagram of the ultrasonic sensor HC-SR04 is shown below:



Piezo Buzzer

In simplest terms, a piezo buzzer is a type of electronic device that's used to produce a tone, alarm or sound. It's lightweight with a simple construction, and it's typically a low-cost product. Yet at the same time, depending on the piezo ceramic buzzer specifications, it's also reliable and can be constructed in a wide range of sizes that work across varying frequencies to produce different sound outputs.

For instance, at [APC International, Ltd.](#), we offer piezo buzzers without signal generators, self-oscillating buzzers that have signal generators and even multi-tone sound generators — often used in alarms and sirens. Regardless of the model you choose, our piezo buzzers offer high sound outputs. Plus, since they can be mounted on circuit boards, they're highly useful in a wide range of applications and assemblies.

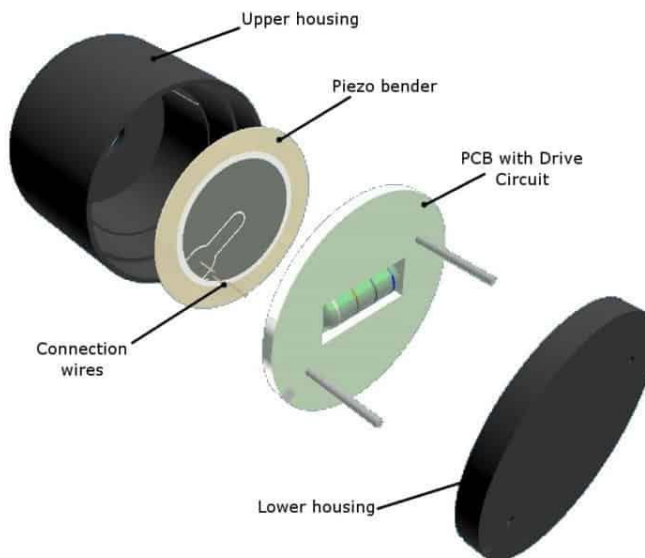
Despite different construction methods that affect the cost of piezo buzzers, all of our prices are highly competitive. In addition, thanks to our state-of-the-art production facility, our delivery times are some of the fastest in the industry.

The piezoelectric buzzer uses the piezoelectric effect of the piezoelectric ceramics and uses the pulse current to drive the vibration of the metal plate to generate sound. Piezoelectric buzzer is

mainly composed of multi-resonator, piezoelectric plate, impedance matcher, resonance box, housing, etc. Some of the piezoelectric buzzers are also equipped with light-emitting diodes. The multi-resonator consists of transistors or integrated circuits. When the power supply is switched on (1.5~15V DC operating voltage), the multi-resonator oscillates and outputs 1.5~2.5 kHz audio signal. The impedance matcher pushes the piezoelectric plate to generate sound. The piezoelectric plate is made of lead zirconate titanate or lead magnesium niobate piezoelectric ceramic, and silver electrodes are plated on both sides of the ceramic sheet. After being polarized and aged, the silver electrodes are bonded together with brass or stainless steel sheets

The piezoelectric buzzer uses the piezoelectric effect of the piezoelectric ceramics and uses the pulse current to drive the vibration of the metal plate to generate sound. Piezoelectric buzzer is mainly composed of multi-resonator, piezoelectric plate, impedance matcher, resonance box, housing, etc. Some of the piezoelectric buzzers are also equipped with light-emitting diodes. The multi-resonator consists of transistors or integrated circuits. When the power supply is switched on (1.5~15V DC operating voltage), the multi-resonator oscillates and outputs 1.5~2.5 kHz audio signal. The impedance matcher pushes the piezoelectric plate to generate sound. The piezoelectric plate is made of lead zirconate titanate or lead magnesium niobate piezoelectric ceramic, and silver electrodes are plated on both sides of the ceramic sheet.

After being polarized and aged, the silver electrodes are bonded together with brass or stainless steel sheets

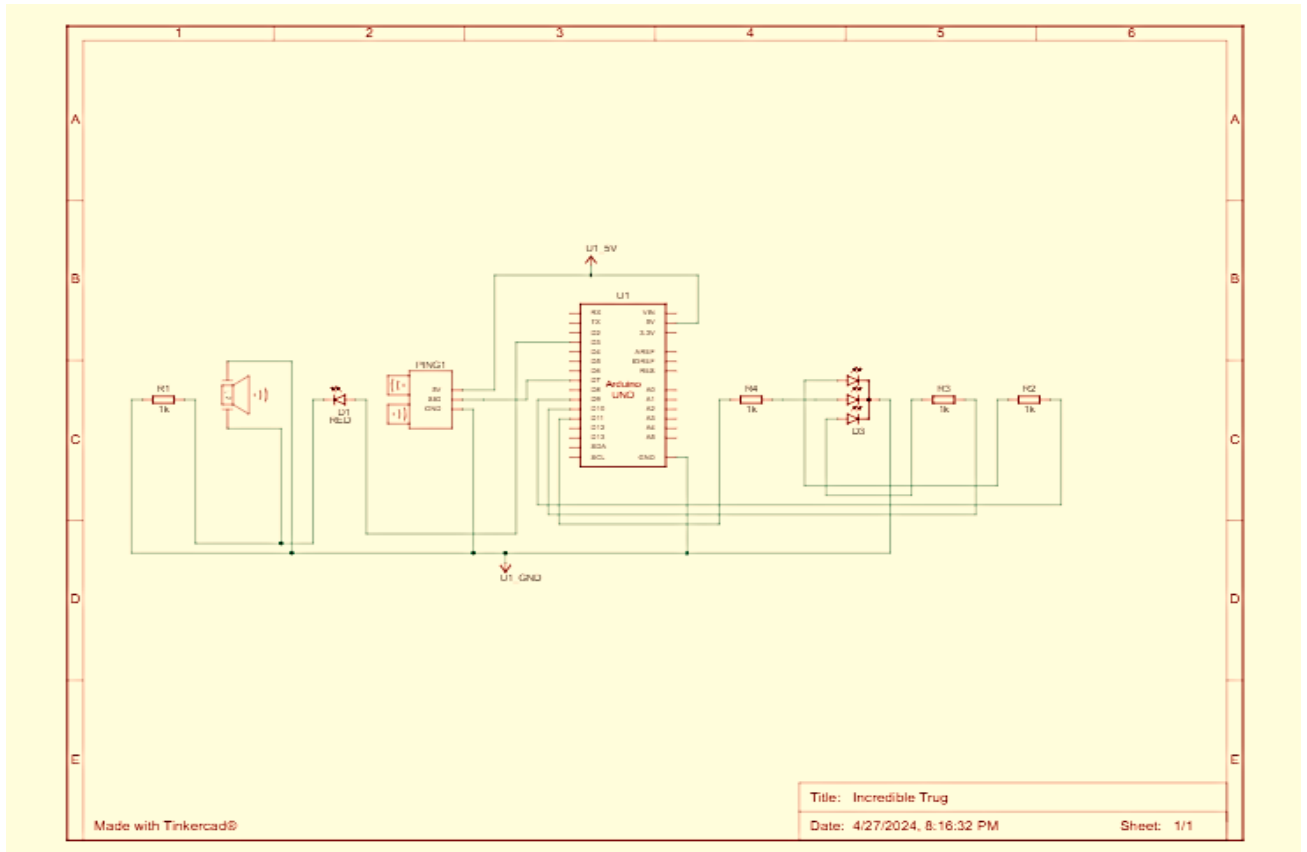


Piezo Buzzer Construction

PROCEDURE

- Connect the 5V pin of PING1 to Power 5V of Arduino board.
- Connect GND pin of PING1 to Power GND of Arduino board.
- Connect SIG pin of PING1 to digital pin 7 of Arduino board.
- Connect LED RGB terminal1 to resistor1, connect terminal2 to GND of Arduino, and connect terminal3 to resistor2 and terminal4 to resistor3.
- Connect resistor1, resistor2, resistor3 to digital pin -11, -10, and 9 respectively.
- Connect Piezo Buzzer terminal1 to GND terminal2 LED terminal1.
- Connect LED terminal1 to resistor4 and connect resistor4 to GND.
- Connect LED terminal2 to digital pin -3 of Arduino.

SCHEMATIC VIEW



COMPONENTS LIST

Name	Quantity	Component
U1	1	Arduino Uno R3
PING1	1	Ultrasonic Distance Sensor
PIEZO1	1	Piezo
D1	1	Red LED
R1 R2 R3 R4	4	1 k Ω Resistor
D3	1	LED RGB

WORKING

- ULTRASONIC Sensor for sound monitoring.
- All the sensors are attached at the ADC pin (analog to digital) to the Arduino microcontroller.
- Microcontroller will do some calculation and send data to IOT module.
- We will set the limit of every sensor if limit exceeded it will alert the by BUZZER.
- IOT module will transmit data so that computer will receive and display data.

CODING

```
const int pingPin = 7;

const int red=11;

const int blue=10;

int green=9;

void setup() {
  // initialize serial communication:
  Serial.begin(9600);
  pinMode(red,OUTPUT);
  pinMode(blue,OUTPUT);
  pinMode(green,OUTPUT);
  pinMode(3, OUTPUT);
}

void loop()
{
  digitalWrite(3, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(3, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
  // establish variables for duration of the ping, and the distance result
  // in inches and centimeters:
  long duration, inches, cm;
  // The PING))) is triggered by a HIGH pulse of 2 or more microseconds.
  // Give a short LOW pulse beforehand to ensure a clean HIGH pulse:
  pinMode(pingPin, OUTPUT);
  digitalWrite(pingPin, LOW);
  delayMicroseconds(2);
```

```

digitalWrite(pingPin, HIGH);
delayMicroseconds(5);
digitalWrite(pingPin, LOW);

// The same pin is used to read the signal from the PING))) a HIGH pulse
// whose duration is the time (in microseconds) from the sending of the ping
// to the reception of its echo off of an object.
pinMode(pingPin, INPUT);
duration = pulseIn(pingPin, HIGH);

// convert the time into a distance
inches = microsecondsToInches(duration);
cm = microsecondsToCentimeters(duration);
Serial.print(inches);
Serial.print("in, ");
Serial.print(cm);
Serial.print("cm");
Serial.println();

if(cm<256){
  analogWrite(red,cm);
  analogWrite(blue,255-cm);
  analogWrite(green,inches);
}
else{
  analogWrite(red,0);
  analogWrite(blue,0);

```



```

    analogWrite(green,0);}

    delay(100);

}

```

```

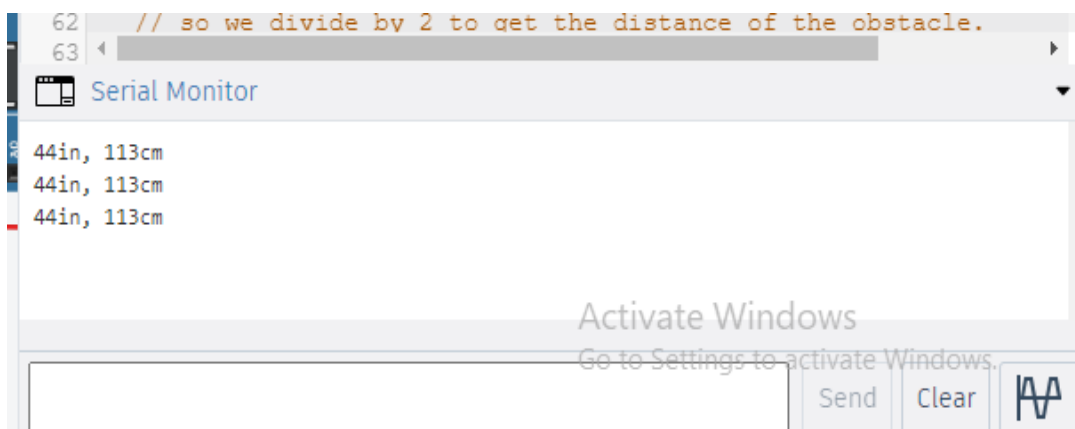
long microsecondsToInches(long microseconds) {
    // According to Parallax's datasheet for the PING))) , there are 73.746
    // microseconds per inch (i.e. sound travels at 1130 feet per second).
    // This gives the distance travelled by the ping, outbound and return,
    // so we divide by 2 to get the distance of the obstacle.
    // See: http://www.parallax.com/dl/docs/prod/acc/28015-PING-v1.3.pdf
    return microseconds / 74 / 2;
}

```

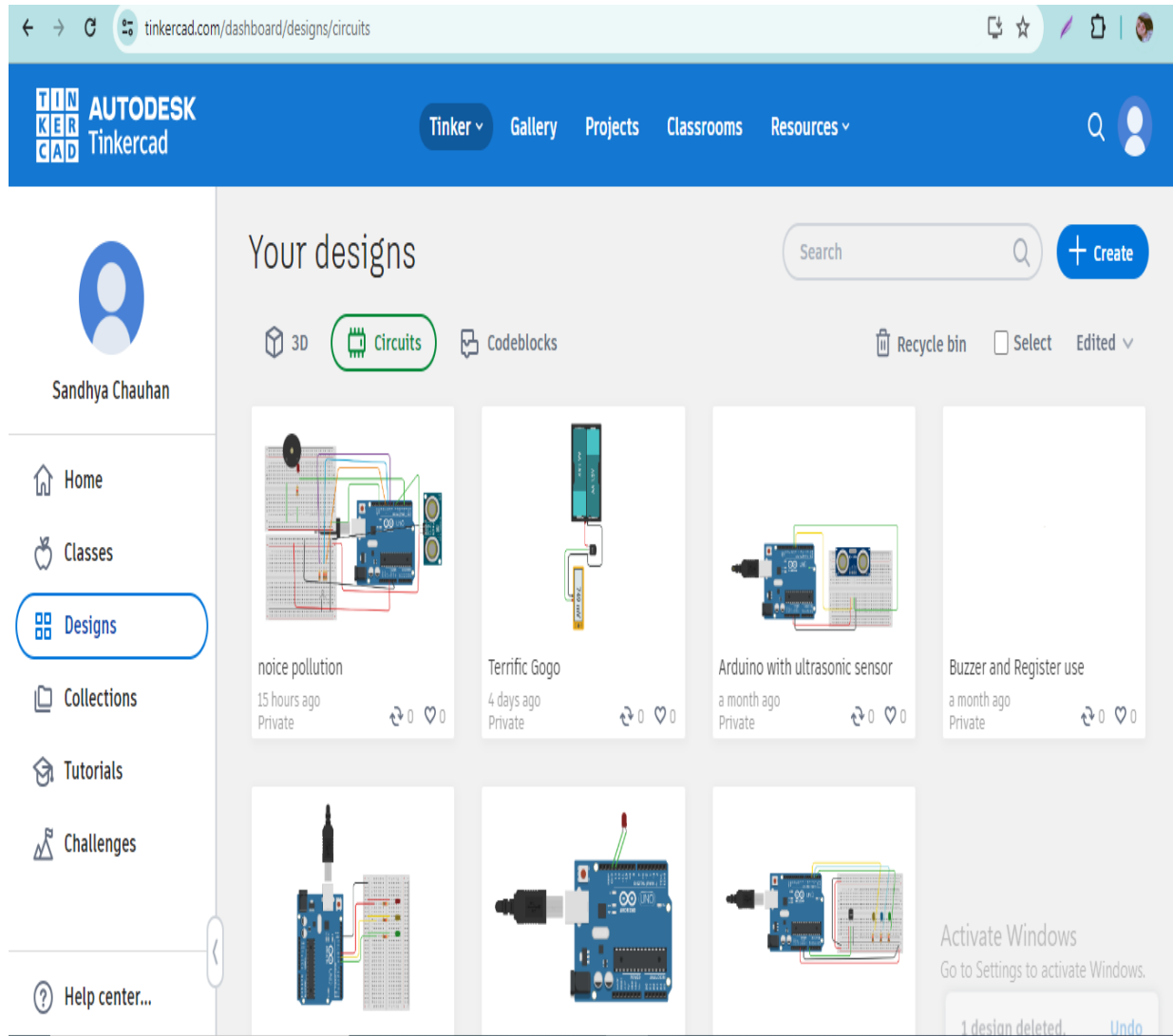
```

long microsecondsToCentimeters(long microseconds) {
    // The speed of sound is 340 m/s or 29 microseconds per centimeter.
    // The ping travels out and back, so to find the distance of the object we
    // take half of the distance travelled.
    return microseconds / 29 / 2;
}

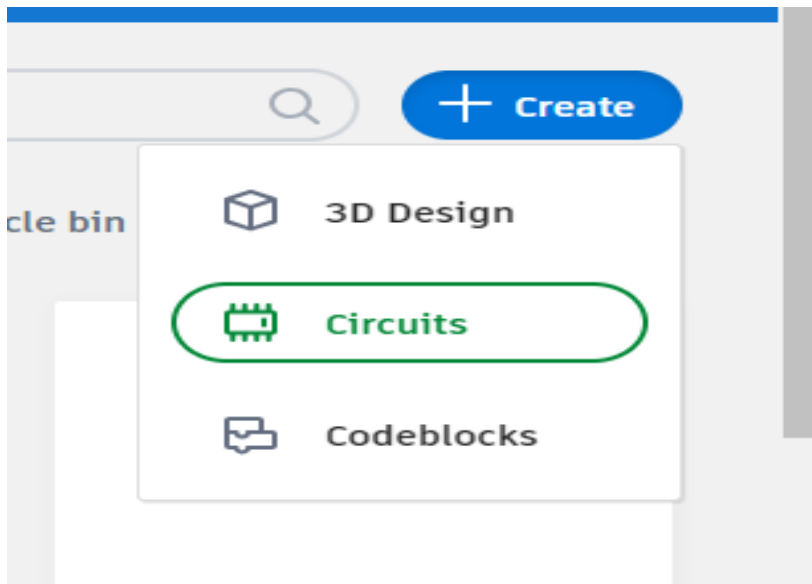
```



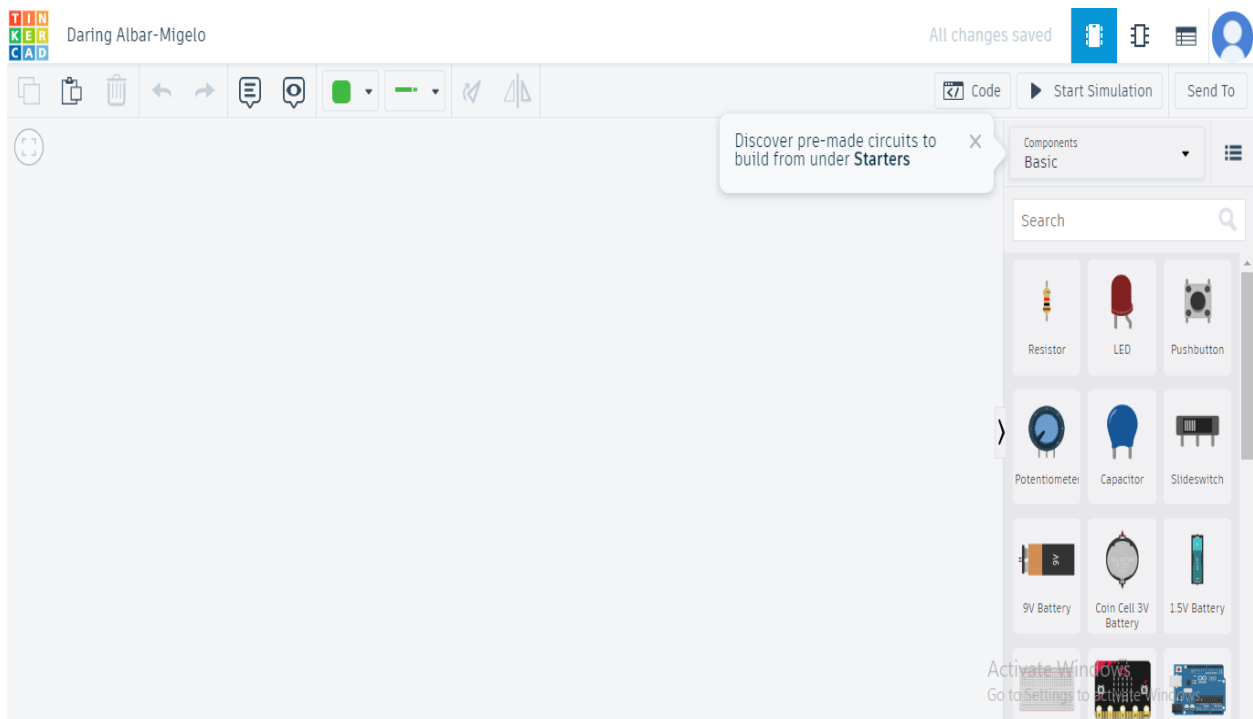
BUILDING A TINKERCAD CIRCUIT



- **Open TINKERCAD AUTODESK .**
- **Click on “+CREATE” for building a Circuit in top right corner.**



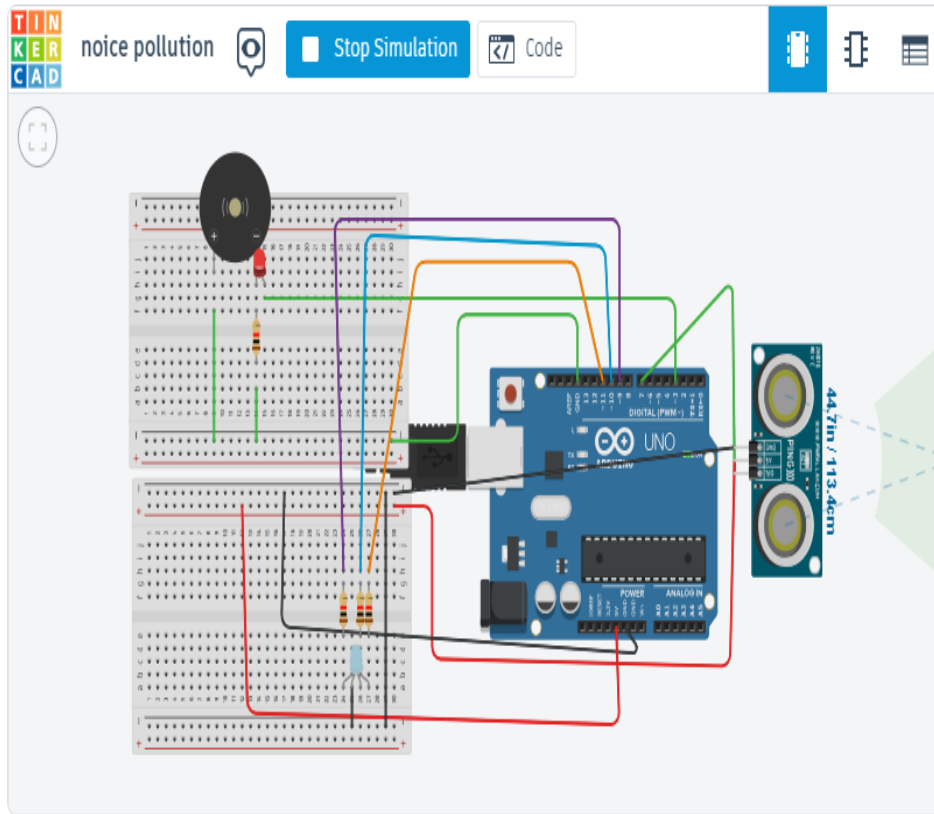
- Now click on “Circuits” to build a circuit.



- You can choose different components for circuit.

noice pollution

React  0



Circuit by
Sandhya Chauhan

Tinker this

 Share to Classroom

Copy link

Design is visible only to you.

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Created April 27, 2024



 Report content

- Click on “Start Stimulation” for execution of circuit and click on “Stop Stimulation” for stopping the execution.
- By clicking on Code you can edit and write your on code.

Code

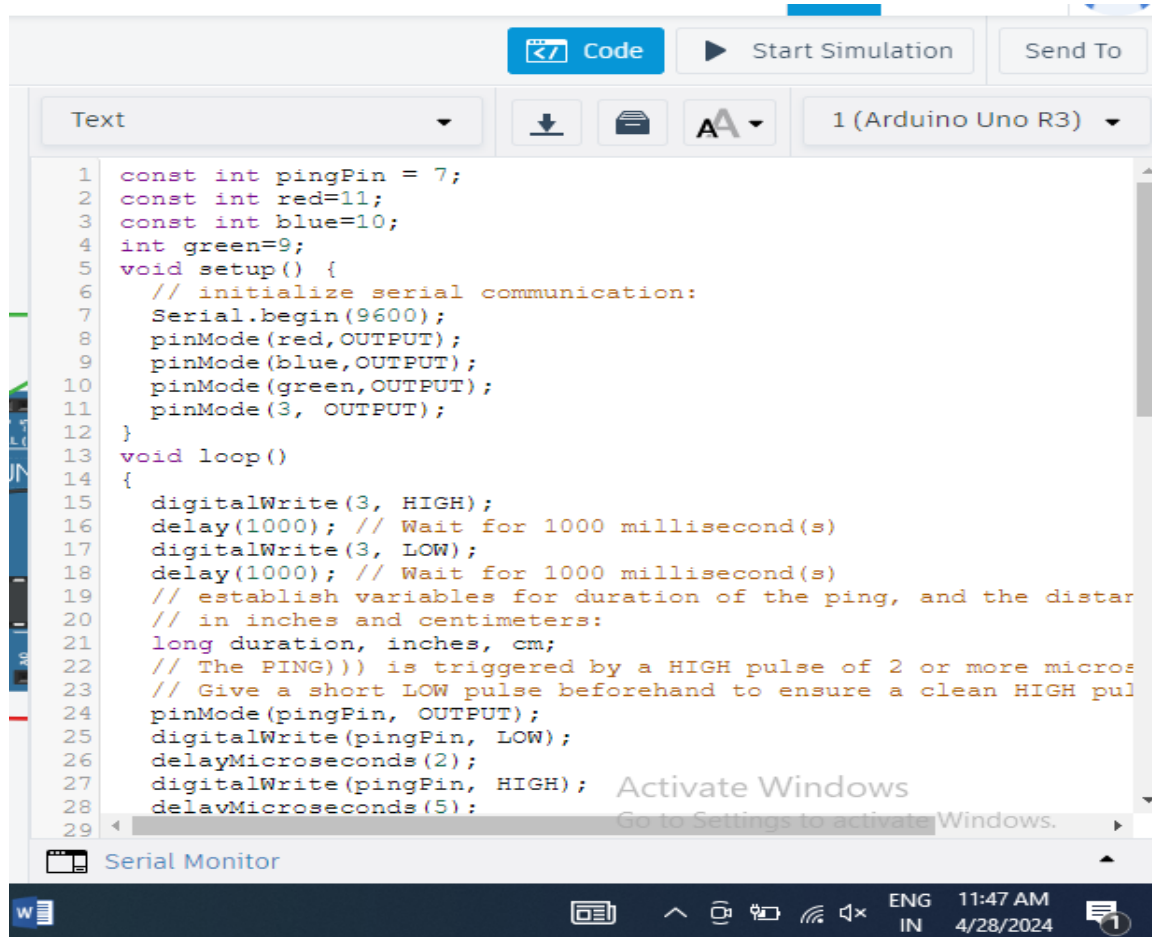
Start Sim

Toggle code editor

```

1  const int pingPin = 7;
2  const int red=11;
3  const int blue=10;
4  int green=9;
5  void setup() {
6    // initialize serial communication:
7    Serial.begin(9600);
8    pinMode(red,OUTPUT);
9    pinMode(blue,OUTPUT);
10   pinMode(green,OUTPUT);
11   pinMode(3, OUTPUT);
12 }
13 void loop()
14 {
15   digitalWrite(3, HIGH);
16   delay(1000); // Wait for 1000 millisecond(s)

```



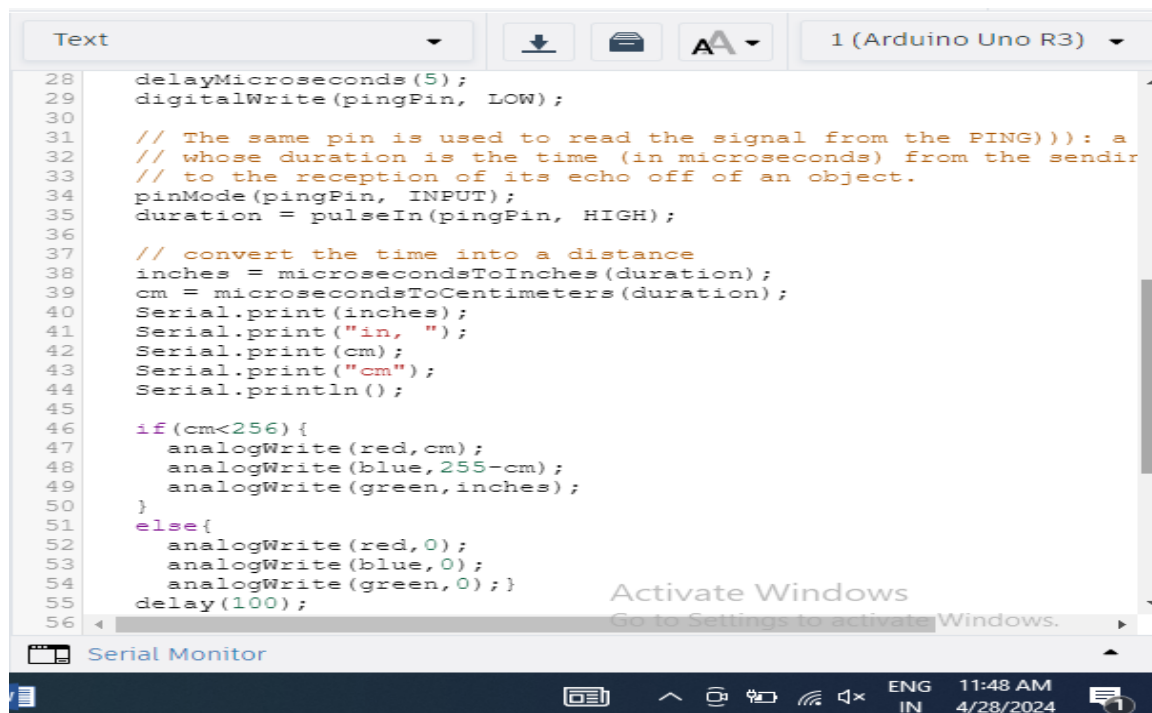
Code editor showing the setup and loop functions for an Arduino Uno R3. The code is for a distance-measuring circuit using a PING sensor and three LEDs (red, blue, green).

```

1  const int pingPin = 7;
2  const int red=11;
3  const int blue=10;
4  int green=9;
5  void setup() {
6      // initialize serial communication:
7      Serial.begin(9600);
8      pinMode(red,OUTPUT);
9      pinMode(blue,OUTPUT);
10     pinMode(green,OUTPUT);
11     pinMode(3, OUTPUT);
12 }
13 void loop()
14 {
15     digitalWrite(3, HIGH);
16     delay(1000); // Wait for 1000 millisecond(s)
17     digitalWrite(3, LOW);
18     delay(1000); // Wait for 1000 millisecond(s)
19     // establish variables for duration of the ping, and the distance
20     // in inches and centimeters:
21     long duration, inches, cm;
22     // The PING))) is triggered by a HIGH pulse of 2 or more microseconds
23     // Give a short LOW pulse beforehand to ensure a clean HIGH pulse
24     pinMode(pingPin, OUTPUT);
25     digitalWrite(pingPin, LOW);
26     delayMicroseconds(2);
27     digitalWrite(pingPin, HIGH);
28     delayMicroseconds(5);

```

Serial Monitor



Code editor showing the continuation of the loop function for the Arduino Uno R3. The code calculates the distance from the ping duration, prints it in inches and centimeters, and controls the LEDs based on the distance.

```

28     delayMicroseconds(5);
29     digitalWrite(pingPin, LOW);
30
31     // The same pin is used to read the signal from the PING))) : a
32     // whose duration is the time (in microseconds) from the sending
33     // to the reception of its echo off of an object.
34     pinMode(pingPin, INPUT);
35     duration = pulseIn(pingPin, HIGH);
36
37     // convert the time into a distance
38     inches = microsecondsToInches(duration);
39     cm = microsecondsToCentimeters(duration);
40     Serial.print(inches);
41     Serial.print("in, ");
42     Serial.print(cm);
43     Serial.print("cm");
44     Serial.println();
45
46     if (cm<256) {
47         analogWrite(red, cm);
48         analogWrite(blue, 255-cm);
49         analogWrite(green, inches);
50     }
51     else {
52         analogWrite(red, 0);
53         analogWrite(blue, 0);
54         analogWrite(green, 0);
55     }
56     delay(100);

```

Serial Monitor

```

45
46   if (cm < 256) {
47       analogWrite (red, cm);
48       analogWrite (blue, 255 - cm);
49       analogWrite (green, inches);
50   }
51   else {
52       analogWrite (red, 0);
53       analogWrite (blue, 0);
54       analogWrite (green, 0);
55   }
56   delay (100);
57 }
58
59 long microsecondsToInches (long microseconds) {
60     // According to Parallax's datasheet for the PING))) , there are
61     // microseconds per inch (i.e. sound travels at 1130 feet per s
62     // This gives the distance travelled by the ping, outbound and
63     // so we divide by 2 to get the distance of the obstacle.
64     // See: http://www.parallax.com/dl/docs/prod/acc/28015-PING-v1.
65     return microseconds / 74 / 2;
66 }
67
68 long microsecondsToCentimeters (long microseconds) {
69     // The speed of sound is 340 m/s or 29 microseconds per centime
70     // The ping travels out and back, so to find the distance of th
71     // take half of the distance travelled.
72     return microseconds / 29 / 2;

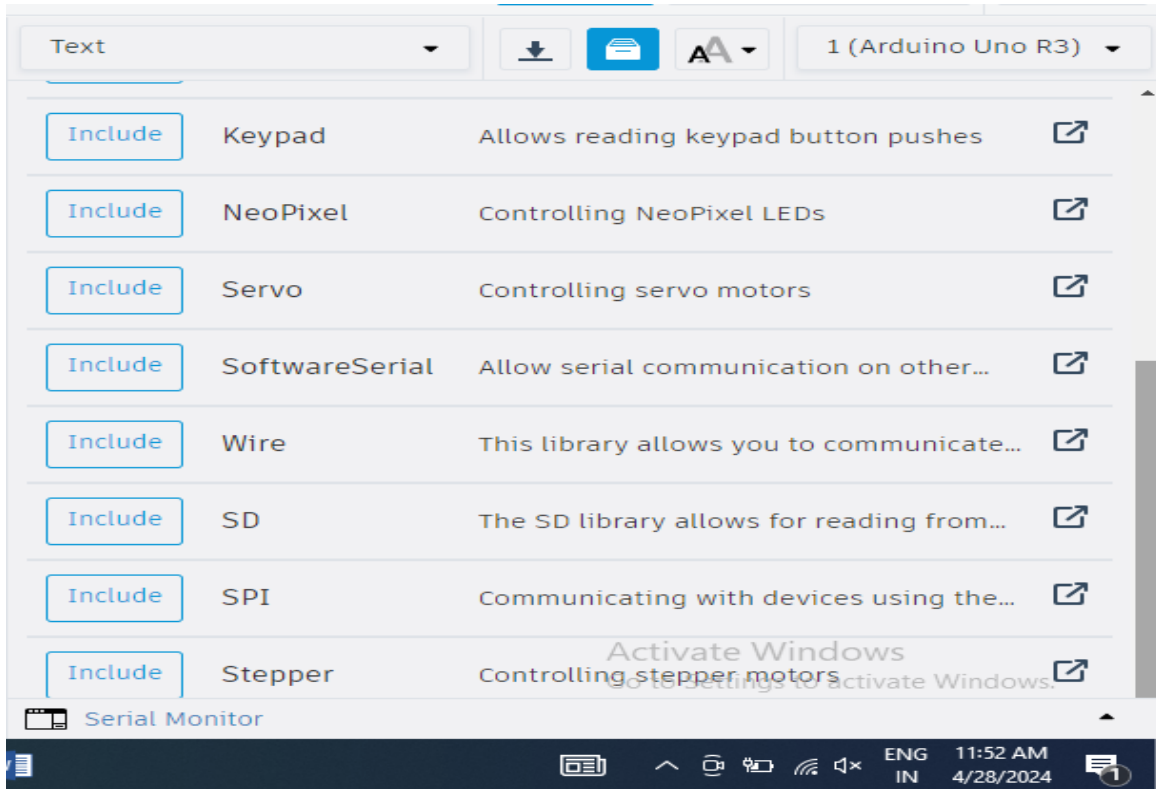
```

Serial Monitor

LIBRARIES USED

Include	Library Name	Description	Link
Include	Adafruit LED Backpack	Library for 8x8 matrix and 7-segment LE...	
Include	Adafruit LiquidCrystal	Controlling liquid crystal displays using...	
Include	EEPROM	Reading and writing to "permanent"...	
Include	IRremote	Library to decode IR sensors	
Include	LiquidCrystal	Controlling liquid crystal displays (LCDs)	
Include	LiquidCrystal I2C	Controlling liquid crystal displays using...	
Include	Keypad	Allows reading keypad button pushes	
Include	NeoPixel	Controlling NeoPixel LEDs	

Serial Monitor



ADVANTAGES AND DISADVANTAGES

Advantages

- The system can be used to monitor sound and can measure distance
- Data can be used in researches and data mining.
- It is easy to use, safe, and reliable.

Disadvantages

- The device is not water resistant; keep away from moisture.
- System may not work during extreme weather conditions of IoT.

FUTURE SCOPE

- In future we will add more accurate sensors for accurate analysis of data.
- We will work on the power supply and replace existing system with solar based power supply.
- In future we modify the system to notify a user about the air quality and noise level it reaches beyond permissible level through sms or app.
- We can monitor air and sound pollution level at any place of the world.
- By using this project each and every variation we can analyze and inform nearby people in time. We can also analyze data from home using thingspeak
- The most important factor of this system is that it is small, cost efficient and portable. Sensors are available easily anywhere. This system is highly helpful to save the lives and overcome all the problems related to environment

REFERENCES

- www.Google.com
- <https://www.researchgate.net>
- <https://www.scribd.com>
- <https://www.tinkercad.com>
- <https://www.javatpoint.com>
- <https://www.arduino.cc>
- “THE INTERNET OF THINGS IN THE CLOUD” by Honbo Zhou.

THANK YOU